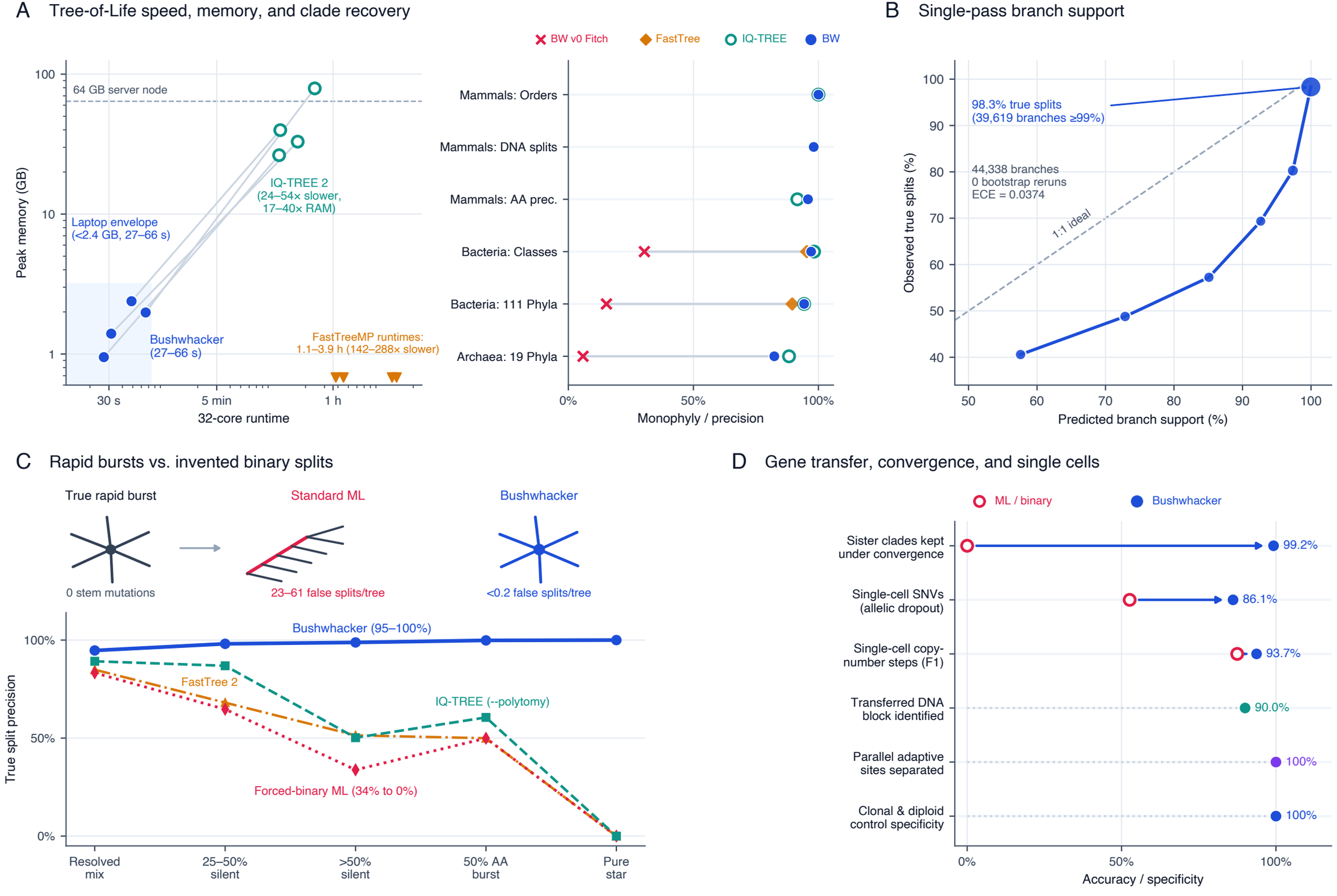
A
Tree-of-Life speed, memory, and clade recovery
Peak memory (GB)
100
10
1
64 GB server node
IQ-TREE 2
(24–54× slower,
17–40× RAM)
Laptop envelope
(<2.4 GB, 27–66 s)
Bushwhacker
(27–66 s)
FastTreeMP runtimes:
1.1–3.9 h (142–288× slower)
30 s
5 min
1 h
32-core runtime
BW v0 Fitch
FastTree
IQ-TREE
BW
Mammals: Orders
Mammals: DNA splits
Mammals: AA prec.
Bacteria: Classes
Bacteria: 111 Phyla
Archaea: 19 Phyla
0%
50%
100%
Monophyly / precision
B
Single-pass branch support
Observed true splits (%)
100
90
80
70
60
50
40
98.3% true splits
(39,619 branches ≥99%)
44,338 branches
0 bootstrap reruns
ECE = 0.0374
1:1 ideal
50
60
70
80
90
100
Predicted branch support (%)
C
Rapid bursts vs. invented binary splits
True rapid burst
Standard ML
Bushwhacker
0 stem mutations
23–61 false splits/tree
<0.2 false splits/tree
Bushwhacker (95–100%)
FastTree 2
IQ-TREE (--polytomy)
Forced-binary ML (34% to 0%)
True split precision
100%
50%
0%
Resolved
mix
25–50%
silent
>50%
silent
50% AA
burst
Pure
star
D
Gene transfer, convergence, and single cells
ML / binary
Bushwhacker
Sister clades kept
under convergence
99.2%
Single-cell SNVs
(allelic dropout)
86.1%
Single-cell copy-
number steps (F1)
93.7%
Transferred DNA
block identified
90.0%
Parallel adaptive
sites separated
100%
Clonal & diploid
control specificity
100%
0%
50%
100%
Accuracy / specificity